Matière: Analyse : المادة
Date: : التاريخ
Semestre : : الفصل
Année Universitaire : 20 ... 20 : السنة

Ex1

1) $\frac{x^2-1}{2x-1} = ax+b+\frac{c}{2x-1}$

$ax+b+\frac{c}{2x-1} = \frac{ax(2x-1)+b(2x-1)+c}{2x-1}$

$= \frac{2ax^2-ax+2bx-b+c}{2x-1}$

$= \frac{2ax^2+(2b-a)x+c-b}{2x-1}$

par l'identification

$\begin{cases} 2a=1 \Rightarrow \boxed{a=\frac{1}{2}} \\ 2b-a=0 \Rightarrow \boxed{b=\frac{1}{4}} \\ c-b=-1 \Rightarrow c=\frac{-3}{4} \end{cases}$

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$\Rightarrow \frac{x^2-1}{2x-1} = \frac{1}{2}x+\frac{1}{4}+\frac{-3/4}{2x-1}$

2) $\int_{-1}^{0} \frac{x^2-1}{2x-1} = \int \frac{1}{2}x+\frac{1}{4}+\frac{-3/4}{2x-1}$

$= \int_{-1}^{0}\frac{1}{2}x + \int_{-1}^{0}\frac{1}{4} + \frac{-3}{8}\int_{-1}^{0}\frac{-2}{1-2x}$

$\left[\frac{x^2}{4}\right]_{-1}^{0} + \left[\frac{1}{4}x\right]_{-1}^{0} - \frac{3}{8}\left[\ln(1-2x)\right]_{-1}^{0}$

$= -\frac{1}{4}+\frac{1}{4} - \frac{3}{8}(-\ln(3)) = \boxed{\frac{3}{8}\ln(3)}$

3) $\int_{-\frac{\pi}{6}}^{0} \frac{\cos^3(x)}{1-2\sin x} = \int_{-\frac{\pi}{6}}^{0} \frac{\cos(x)\cdot\cos^2(x)}{1-2\sin(x)}$

on a : $\sin^2+\cos^2(x)=1$

$\Rightarrow \cos^2(x) = 1-\sin^2(x)$

on pose : $u=\sin(x)$ | $x=0,\ u=0$

$du = \cos(x)\,dx$ | $x=-\frac{\pi}{6},\ u=-\frac{1}{2}$

$dx = \frac{1}{\cos(x)}du$

$\int_{-\frac{\pi}{6}}^{0} \frac{\cancel{\cos(x)}(1-\sin^2(x))\times\frac{1}{\cancel{\cos(x)}}\,du}{1-2\sin(x)}$

$= \int_{-\frac{\pi}{6}}^{0} \frac{1-u^2}{1-2u}du = \int_{-\frac{\pi}{6}}^{0}\frac{u^2-1}{2u-1}$

$$\int_{-\frac{1}{2}}^{0} \frac{1}{2}u+\frac{1}{4}+\frac{-\frac{3}{4}}{2u-1}$$

$$\left[\cancel{\frac{u^2}{4}}\right]_{-\frac{1}{2}}^{0}+\left[\cancel{\frac{1}{4}u}\right]_{-\frac{1}{2}}^{0}-\frac{3}{8}\left[\ln|2u-1|\right]$$

$$\int_{-\frac{1}{2}}^{0} \frac{1}{2}u+\frac{1}{4}-\frac{3}{8}\times\frac{-2}{1-2u}$$

$$\left[\frac{1}{4}u^2\right]_{-\frac{1}{2}}^{0}+\left[\frac{1}{4}u\right]_{-\frac{1}{2}}^{0}-\frac{3}{8}\left[\ln|1-2u|\right]_{-\frac{1}{2}}^{0}$$

$$-\frac{1}{16}+\frac{1}{8}-\frac{3}{8}(-\ln(2))$$

$$=\frac{1}{16}+\frac{3}{8}\ln(2)$$

Ex2:

$$\int_1^e \frac{\ln(x)}{x}dx$$

on pose $u=\ln(x)$

$$du=\frac{1}{x}dx$$

$$\Rightarrow dx=x\,du$$

si $x=e$ $u=1$

si $x=1$ $u=0$

$$\int_0^1 \frac{u}{\cancel{x}}\times\cancel{x}\,du=\int_0^1 u\,du$$

$$\left[\frac{u^2}{2}\right]_0^1=\frac{1}{2}$$

2) $\displaystyle\int_0^{\frac{\pi}{2}} \sin x\cos^2(x)\,dx$

on pose $u=\cos(x)$

$$du=-\sin(x)dx$$

$$dx=-\frac{1}{\sin(x)}du$$

si $x=\frac{\pi}{2}$ $u=0$

$x=0$ $u=1$

$$-\int_0^1 \cancel{\sin x}\times u^2\times-\frac{1}{\cancel{\sin x}}du$$

$$=\int_0^1 u^2=\left[\frac{u^3}{3}\right]_0^1=\boxed{\frac{1}{3}}$$

3) $\displaystyle\int_0^1 \sqrt{1-x^2}\,dx$

on pose $x=\cos(u)$

$$dx=-\sin(u)du$$

si $x=1$ $u=0$

si $x=0$ $u=\frac{\pi}{2}$

$$-\int_0^{\frac{\pi}{2}} \sqrt{1-\cos^2(u)}\times-\sin(u)du$$

on a

$$\cos^2(u)+\sin^2(u)=1$$

$$\Rightarrow \sin(u)=\sqrt{1-\cos^2(u)}$$

$$\int_0^{\frac{\pi}{2}} \sin^2(u)=\left[\frac{2u-\sin 2u}{4}\right]_0^{\frac{\pi}{2}}$$

$$=\frac{\pi}{4}$$

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Ex 3 :

1) $\displaystyle\int_a^t \frac{1}{5+x^2}\,dx = \int_a^t \frac{1}{(\sqrt5)^2+x^2}$

$= \left[\frac{1}{\sqrt5}\,\text{Artan}\left(\frac{x}{\sqrt5}\right)\right]_a^t$

$= \frac{1}{\sqrt5}\,\text{Artan}\left(\frac{t}{\sqrt5}\right) + C$

2) $\displaystyle\int_a^t \frac{1}{\sqrt{x^2-5}} = \left[\ln(x+\sqrt{x^2-5})\right]_a^t$

$= \ln|t+\sqrt{t^2-5}| + C$

3) $\displaystyle\int_a^t \frac{\ln(x)}{x} = \int_a^t \frac{1}{x}\times \ln(x)$

$= \left[\frac{\ln(x)^2}{2}\right]_a^t = \frac{\ln(t)^2}{2} + C$

4) $\displaystyle\int_a^t \cos^3(x)\,dx = \int_a^t \cos(x)\cdot\cos^2(x)\,dx$

$\displaystyle\int_a^t \cos(x)\,(1-\sin^2(x))$

$= \displaystyle\int_a^t \cos(x) - \cos(x)\sin^2(x)$

$= \left[\sin(x) - \frac{\sin^3}{3}\right]_a^t$

$= \sin(t) - \frac{\sin(t)^3}{3} + C$

Ex 4 :

1) $\displaystyle\int_0^1 \frac{\text{Arctan}(x)}{1+x^2} = \int_0^1 \frac{1}{1+x^2}\times \text{Artan}(x)$

$\left[\frac{\text{Artan}^2(x)}{2}\right]_0^1 = \frac{\text{Artan}^2(1)}{2}$

2°) $\displaystyle\int_1^2 x^2\ln(x)\,dx$

on pose

$u = \ln(x)$, $u' = \frac{1}{x}$

$v' = x^2$, $v = \frac{x^3}{3}$

$\displaystyle\int_1^2 x^2\ln(x) = \left[\frac{x^3}{3}\ln(x)\right]_1^2 - \int \frac{1}{x}\times\frac{x\times x^2}{3}$

$= \left[\frac{x^3}{3}\ln(x)\right]_1^2 - \frac{1}{3}\left[\frac{x^3}{3}\right]_1^2$

$= \frac{8}{3}\ln(2) - \frac{8}{9} + \frac{1}{9} =$

$\frac{8}{3}\ln(2) - \frac{7}{9}$

3) $\displaystyle\int_0^1 \frac{3x+1}{(x+1)^2} = \frac{3x+3-3+1}{(x+1)^2}$

$= \displaystyle\int_0^1 \frac{3(x+1)}{(x+1)^2} - \frac{2}{(x+1)^2} = \int_0^1 \frac{3}{x+1} - \frac{2}{(x+1)^2}$

$= 3\displaystyle\int_0^1 \frac{1}{x+1} - 2\int_0^1 \frac{1}{x+1}$

$\cancel{\int_0^1 3\ln(x+1) + \int_0^1 2\ln(x+1)}$

$= \cancel{\int_0^1 -\ln(x+1)}$

$3\left[\ln(x+1)\right]_0^1 + 2\left[\frac{1}{x+1}\right]_0^1$

$= 3\ln(2) + 2\left(\frac{1}{2} - 1\right) = 3\ln(2) - 1$

$\displaystyle \int \frac{1}{(1+x^2)^2}\,dx$

on pose $x=\tan(u)$

$dx=(1+\tan^2(u))\,du$

si $x=1$ $\quad u=\frac{\pi}{4}$

si $x=0$ $\quad u=0$

$$\int_0^{\frac{\pi}{4}} \frac{1}{(1+(\tan(u))^2)^2} \times (1+\tan^2(u))\,du$$

$$=\int_0^{\frac{\pi}{4}} \frac{1}{1+\tan^2(u)} = \int_0^{\frac{\pi}{4}} \frac{1}{1+\frac{\sin^2(u)}{\cos^2(u)}}$$

$$\int_0^{\frac{\pi}{4}} \frac{1}{\frac{\cos^2+\sin^2(u)}{\cos^2(u)}} = \int_0^{\frac{\pi}{4}} \cos^2(u)$$

$$=\left[\frac{2u+\sin(2u)}{4}\right]_0^{\frac{\pi}{4}}$$

$$\frac{\frac{\pi}{2}+1}{4} = \boxed{\frac{\pi+2}{8}}$$